Non-linear heat conduction

Material properties are T-dependent

PDE: $-\nabla\cdot[k(T)\nabla T] = f_s$

$\Rightarrow$ non-linear

Example: Geotherm of Europa's ice shell

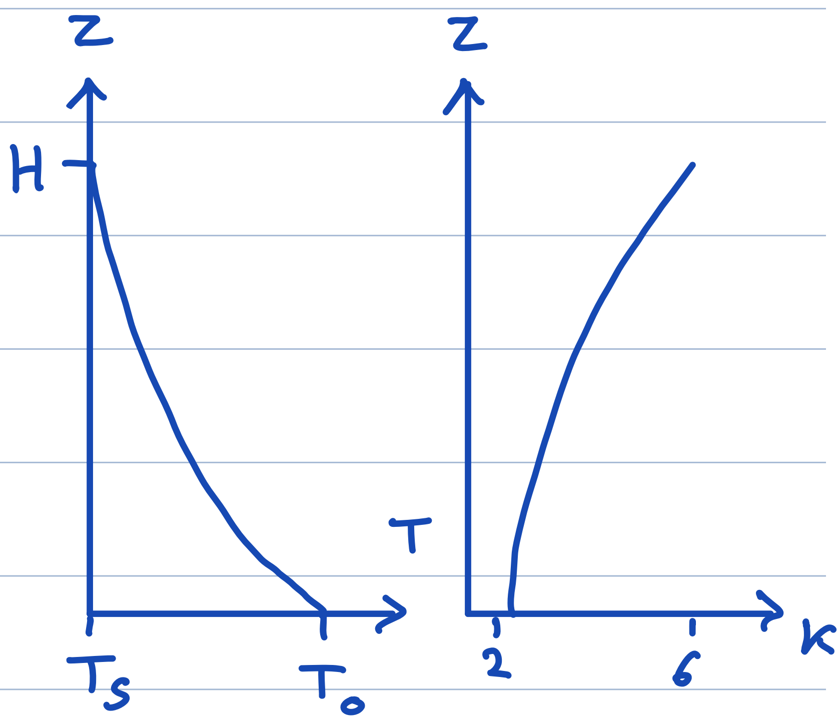
ice: $\boxed{k = \frac{612}{T}}$ $\left[\frac{W}{m\,K}\right]$ (Carnahan et al. 2021)

PDE: $-\nabla\cdot[k(T)\nabla T] = 0$ $\quad z \in [0, H]$

BC: $T(0) = 273\text{ K}$ $\quad T(H) = 100\text{ K}$

Analytic solution:

$$\boxed{T(z) = T_0\, e^{\ln\left(\frac{T_s}{T_0}\right)\frac{z}{H}}}$$



Discretizing non-linear equations

Continuum: $-\nabla \cdot [\kappa(T)\nabla T] = f_s$

Discrete: $-\underline{\underline{D}} * \underline{\underline{Kd}}(\underline{u}) * \underline{\underline{G}}\,\underline{u} = \underline{f_s}$

Discrete problem → non-linear algebraic system

$$\underline{N}(\underline{u}) = \underline{f_s}$$

$\underline{N}(\underline{u})$ non-linear vector-valued vector function

Note: cannot be written $\underline{\underline{L}}\,\underline{u} = \underline{f_s}$!

Q1: How do we compute $\underline{\underline{Kd}}$?

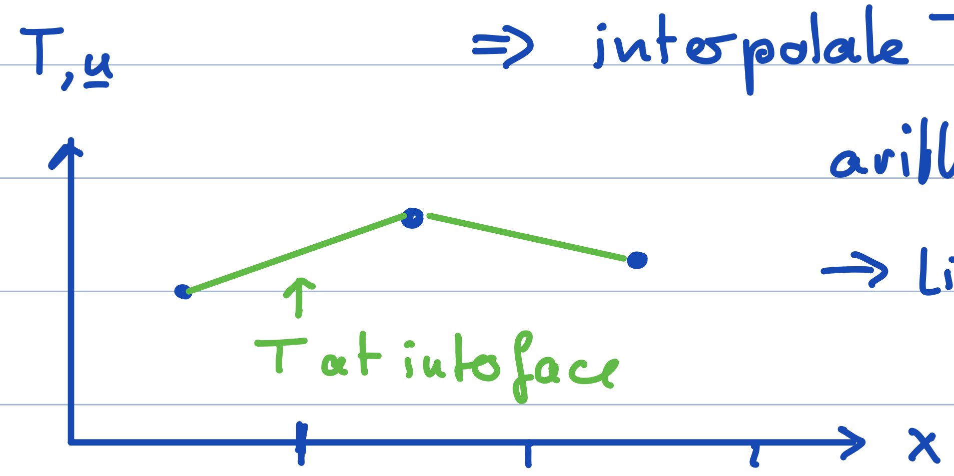
Q2: How do we solve $\underline{N}(\underline{u}) = \underline{f_s}$

$\kappa = \kappa(T)$ and T is smooth

$\Rightarrow$ interpolate T to face

arithmetic mean

→ Linear interpolation



Q1: Which average for $\underline{\underline{k_d}}$?

$\underline{\text{kmean}}$ Nf by 1 vector of face averages

a) Average T then evaluate κ

$\underline{\text{kmean}} = \underline{\kappa}(\underline{\underline{M}}\,\underline{u})$

b) Evaluate κ then average

$\underline{\text{kmean}} = \underline{\underline{M}}\;\underline{\kappa}(\underline{u})$

↳ for consistency with a)

⇒ for non-liner coefficients we use arithmetic mean (and typically option b)

Q2: Newton-Raphson Method !